

# Segmentation of Medical Images Using Deep Learning and Texture Enhancement Based on Fractional Derivative Operators

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**Abstract**—This paper explores the task of automatic intracranial hemorrhage (ICH) segmentation based on computer tomography (CT) data, with potential applications in biomedical engineering. The main focus is, on improving segmentation accuracy by incorporating texture enhancement techniques based on fractional order derivatives. The study looks at the segmentation of ICH using U-Net, a deep learning model that is widely used in the field of the segmentation of medical images. The training process employs a parallel algorithm using CUDA technology. Afterwards, an investigation is conducted into a texture enhancement technique that relies on Riesz fractional order derivatives. The goal is to capture intricate details and subtle textures with the potential to enhance segmentation accuracy. To assess the impact of this preprocessing method on the automatic ICH segmentation problem, the U-Net model undergoes retraining and validation. The texture-enhanced images are analyzed to interpret the obtained results. The findings indicate a subtle yet discernible enhancement in accuracy, as gauged by the Jaccard and Dice coefficients. This emphasizes the auspicious potential of the explored texture enhancement method in improving intracranial hemorrhage segmentation within the realm of biomedical engineering.

**Keywords**—medical image segmentation, PyTorch framework, fractional order derivative, deep learning, CUDA technology.

## I. INTRODUCTION

Interpreting medical images can be difficult due to their low resolution, high levels of noise, and intricate textures. Despite this, medical images carry a wealth of crucial information. The analysis of these images is further complicated by the presence of noise. Since, it is essential to use texture enhancement and noise reduction techniques to improve the accuracy of medical image analysis.

ANNs (Artificial neural networks) have demonstrated promising results in segmenting medical images. Contemporary ANNs exhibit fairly high accuracy in this task. Nevertheless, there is room for enhancement by employing computer vision algorithms that consider the unique characteristics of the data. Using fractional differential operators to improve the texture of medical images is an innovative approach to applying microelectromechanical systems (MEMS) technology to biomedical imaging. Such approaches can be used in MEMS technologies for various non-optical systems, including computed tomography.

Fractal analysis methods are extensively employed in digital signal processing and have proven to be effective in combating complex noise. As an image can be viewed as a two-dimensional signal, certain methods have been adapted to address computer vision challenges. Numerous image

enhancement techniques based on fractional order derivatives have been suggested. This paper implements one of these methods and examines its impact on the automatic segmentation of intracranial hemorrhages on CT scans using artificial neural networks (ANNs).

## II. RELATED WORKS

### A. The use of Artificial Neural Networks in the Classification of Intracerebral Hemorrhagic Events as Revealed through Computed Tomography Imagery

To date, there has been insufficient study of the segmenting intracranial hemorrhages automatically using CT-scan data. The challenge arises from the scarcity of publicly available datasets with labeled hemorrhage areas, which is a common issue in many medical image processing tasks.

Datasets for classifying CT scans are more commonly available identifying the presence and type of hemorrhage on an image is much more straightforward than pinpointing its location. A CQ500 dataset, which contains 491 head CT scans, is widely used for validating the classifier. It indicates the corresponding type of hemorrhage or its absence. For this data set, successful results were obtained with a recognition accuracy of around 95% for each class. [1]. Furthermore, the Radiological Society of North America (RSNA) recently organised a competition on the Kaggle platform. A dataset of more than 25,000 images has been made available to the public [2].

The abundance of publicly available data has facilitated significant advancements in the classification of brain CT scans for the presence of hemorrhages. Nevertheless, the issue of segmentation remains a significant challenge, primarily due to the lack of suitable standard datasets for this particular task. In their review of contemporary intracranial haemorrhage segmentation techniques, Hsayeni and colleagues highlighted this issue. [3] that both classical computer vision algorithms and deep learning methods are successful in solving this problem. However, proper model comparison is complicated by the absence of standardized datasets for testing.

To tackle this issue, the authors created an open dataset on the PhysioNet platform [5] for validating and training intracranial haemorrhage segmentation models [4]. This study employs the aforementioned dataset.

### B. Analysis of results for selection of data set

In the paper mentioned as reference [3], the authors describe training the U-Net network using a dataset they created. They also assess the accuracy of the model's output using various metrics.

For the training process, we used separate layers that the selected samples were extracted from three-dimensional scans

utilising a window of dimensions WL: 40 and WW: 120. Samples from both patients with and without hemorrhage were used, but due to the imbalance in the number of samples, only a randomly selected portion of the samples without hemorrhage was used. Additionally, to improve the model's generalization, during the training phase, the data underwent a series of random affine transformations. This was necessary due to the wide age range and significant variation in skull sizes, resulting in limited data availability.

Were conducted two experiments. The first employed full images, while the second utilised images that had been divided into 160x160 blocks. The second experiment was given more attention. This model was trained with the Adam optimiser, employing a learning coefficient of 10-5, and the loss function was set to cross-entropy. Training was carried out for 150 epochs with a batch size of 32, using an Nvidia GeForce RTX2080.

Table I displays the results of testing the trained network. The authors have presented the maximum and minimum values, as well as the mean value, of each metric, together with the calculated standard deviations, for all images in the test dataset. The Dice and Jaccard coefficients exhibit relatively modest values, which aligns with observations from prior experiments where the network underwent training on a modestly sized dataset.

TABLE I. RESULTS OF TESTING THE TRAINED ANN [3]

| Test dataset statistics | Dice coefficient | Jaccard coefficient |
|-------------------------|------------------|---------------------|
| Average                 | 0.315            | 0.218               |
| Minimum                 | 0.00             | 0.00                |
| Maximum                 | 0.677            | 0.528               |
| STD                     | 0.211            | 0.163               |

### C. Improving image quality using fractional order derivatives.

The most commonly used techniques for enhancing the quality of an image involve the utilization of integer order derivatives. These derivatives are calculated using various masks such as Sobel, Roberts, or Prewitt masks, which are often referred to as filters, templates, or operators. These operators have certain limitations. The most significant of which is their susceptibility to noise [7].

The frequency of the image determines how much the grey levels change in relation to the distance between pixels. High-frequency components, such as edges and noise, vary a lot in intensity from one pixel to the next. On the other hand, low-frequency components, such as textures and backgrounds, have only slight changes in grey levels. This is according to [8].

It is known that noise has a high frequency. A high-pass filter can enhance an image by making things stand out more. But it may also increase the noise level. However, a method proposed in 2010 by Yui-Fei Pu et al. [9] can overcome this drawback. The method uses fractional order derivatives (YuiFeiPU operators). The authors suggest using the Riemann-Liouville and Grünwald-Letnikov derivatives to make such masks and give several ways to do this. Yui-Fei's mask has 8 parts. For the left and right derivatives along the x-axis, y-axis, and diagonals. This makes the operator rotation-independent.

In a research paper cited as [7], the authors introduced a modified operator called FCD that utilizes the Riesz fractional

derivative in its masks. This modification has shown promising results in enhancing noise resistance. Table II in the paper presents SNR (Signal to Noise Ratio) values obtained for different operators, including the FCD operator with varying mask sizes.

According to the research conducted in [9], a fractional order operator can be beneficial in the processing of CT-scans. Additionally, studies mentioned in [7] and [8] have utilized fractional operators to enhance medical images' texture. However, no research has been conducted to determine the effect of this image enhancement uses ANN for automatic segmentation. Consequently, the objective of this study is to evaluate how the fractional order derivative operator influences the quality of intracranial hemorrhage segmentation using U-Net.

TABLE II. SNR FOR DIFFERENT OPERATORS [7]

| Operator  | SNR  | Mask size | SNR  |
|-----------|------|-----------|------|
| FCD       | 8.0  | 9x9       | 10.2 |
| YiFeiPU   | 8.8  | 7x7       | 8.3  |
| Laplacian | 10.2 | 5x5       | 3.3  |
| Sobel     | 13.3 | 3x3       | 3.5  |

## III. METHODS AND MATERIALS

### A. Intracranial hemorrhage segmentation dataset

The data used to create the dataset was collected at the Al Gira Hospital in Iraq between the months of February and August 2018 [3]. A Siemens/SOMATOM Definition AS computer tomograph was employed to undertake a total of 82 scans on patients, comprising 36 females and 46 males, with an average age of 27.8±19.5 years. The age range of the patients was from 1 day to 72 years, with 27 patients under the age of 18 and 55 over the age of 18. Among the patients, 36 had intracranial hemorrhage, and 22 had skull fractures [3]. The dataset consists of 318 images with existing hemorrhages and 2,491 images without them if we consider individual layers.

The medical images are provided in the NIfTI format and have a resolution of 512x512. The number of layers for each patient is variable, typically comprising approximately 30. Furthermore, each image is accompanied by a corresponding mask. In the event of a haemorrhage, the affected pixels are indicated by a white highlight on the mask.

### B. The U-Net network architecture

Modern versions of U-Net usually differ from the original model. One of these variations was utilized in this study. In this iteration, padding is applied to the input data prior to convolutions, thereby ensuring that the output image is of the same dimensions as the input image. In lieu of inverse convolution, the network decoder employs bilinear interpolation. Furthermore, convolution blocks are equipped with batch normalization and drop-out.

In order to evaluate how well the segmentation process is working, two common metrics are used: Dice and Jaccard coefficients. Jaccard coefficient is calculated by comparing the true image (y) to the image produced by the ANN ( $\hat{y}$ ). It quantifies the proportion of pixels that form the intersection between the selected segment masks, relative to the total number of pixels comprising these segments.

$$J(y, \hat{y}) = \frac{|y \cap \hat{y}|}{|y \cup \hat{y}|} \quad (1)$$

In order to calculate the dice coefficient, it is necessary first to determine the number of pixels that are present in the intersection between two given segments. These numbers are then multiplied by two and then divided by the total number of pixels present in each of the segments under consideration.

$$D(y, \hat{y}) = \frac{2|y \cap \hat{y}|}{|y| + |\hat{y}|} \quad (2)$$

### C. Image quality enhancement algorithm based on the Riesz derivative

This section outlines the primary steps involved in applying the FCD operator [10], with additional approximations using the Riesz operator (3). Another approximations outlined in [11, 12].

The Riesz derivative is the basis for the FCD operator. The formula for the Riesz fractional derivative  $D_R^\nu$  of order  $\nu$  for an infinite interval  $-\infty < x < +\infty$  is as follows:

$$D_R^\nu s(x) = \frac{\partial^\nu s(x)}{\partial |x|^\nu} = -\frac{1}{2\cos(\frac{\pi\nu}{2})} \left( \frac{\partial^\nu s(x)}{\partial x^\nu} + \frac{\partial^\nu s(x)}{\partial (-x)^\nu} \right), \quad (3)$$

where  $\nu \neq 1, p-1, \nu \leq p \leq 2$  for  $p \in \mathbb{N}$  and

$$\frac{\partial^\nu s(x)}{\partial x^\nu} = \frac{1}{\Gamma(p-\nu)} \frac{\partial^p}{\partial x^p} \int_{-\infty}^x \frac{s(\xi) d\xi}{(x-\xi)^{\nu+1-p}} \quad (4)$$

$$\frac{\partial^\nu s(x)}{\partial (-x)^\nu} = \frac{(-1)^p}{\Gamma(\nu-p)} \frac{\partial^p}{\partial x^p} \int_x^{+\infty} \frac{s(\xi) d\xi}{(\xi-x)^{\nu+1-p}} \quad (5)$$

In order to calculate the Riesz derivative of order, the FCD operator algorithm employs a step size of  $h = 1$  through the use of discretisation  $\nu(0 < \nu < 1)$

$$\frac{\partial^\nu s(x)}{\partial |x|^\nu} = -\frac{1}{2\cos(\frac{\pi\nu}{2})h^\nu} 2\omega_0 s(x) [\sum_{k=-\infty}^{+\infty} \omega_k (x - kh)], \quad (6)$$

$$\text{where } \omega_0 = -\frac{\Gamma(1-\nu/2)}{\nu\Gamma(1+\nu/2)\Gamma(-\nu)} \sum_{k=-\infty}^{+\infty} \omega_k^* s(x - kh) \quad (7)$$

$$\omega_k^* = 2\omega_0 \text{ for } k = 1.$$

In order to apply the FCD operator algorithm to an image, three input parameters must be provided based on the Riesz derivative.

- $s$  : the matrix of the image
- $\nu$  : the Riesz derivative order, where  $0 < \nu < 1$
- $m$  : a natural number is employed in order to determine the size of the window. The mask should have a center, so the size of the window will be  $(2m + 1) \times (2m + 1)$

Using the formula resulting from the discretisation (6), and (7), the algorithm starts by calculating the  $C_s$  coefficients.

$$C_{s_k} = -\frac{1}{2\cos(\frac{\pi\nu}{2})h^\nu} \omega_k \quad (8)$$

for  $k = 0, 1, 2, \dots, 2m$ .

With the given coefficients, it is feasible to create eight  $W_l$  masks. The masks are applied to the image via a convolution operation, with the centre point designated as  $C_{s_0}$ . The coefficient  $C_{s_0}$  is the value that corresponds to the current pixel. For this purpose, we need to calculate the value of each

pixel  $s(x, y)$ , where  $x$  represents the row in the  $s$  matrix, and  $y$  represents the column:

$$s_l(x, y) = \sum_{i=M_l}^{N_l} \sum_{j=P_l}^{Q_l} W_l(i, j) s(x + i, y + j), \quad (9)$$

for  $l = 1, 2, 3, 4$ ;

$$s_l(x, y) = \sum_{i=M_l}^{N_l} \sum_{j=P_l}^{Q_l} 2^{-\nu/2} W_l(i, j) s(x + i, y + j) + (1 - 2^{-\nu/2}) W_l(0, 0) s(x, y), \quad (10)$$

for  $l = 5, 6, 7, 8$ ;

where

$$\begin{array}{llll} M_1 = M_6 = M_7 = -2m & N_1 = N_6 = N_7 = 0 & P_2 = P_3 = P_7 = -2m & Q_1 = Q_2 = m \\ M_5 = M_4 = -m & N_2 = N_5 = N_8 = 2m & P_1 = -2m & Q_3 = Q_5 = Q_7 = 0 \\ M_5 = M_8 = 0 & N_3 = N_4 = m & P_4 = P_6 = P_8 = 0 & Q_4 = Q_6 = Q_8 = 2m \\ & & P_5 = -2 & \end{array}$$

Subsequently, the final value is calculated by aggregating the results obtained from the application of all the masks.

$$s(x, y) = \frac{\sum_{l=1}^8 s_l(x, y)}{4 \left[ \sum_{k=0}^n C_{s_k} + \sum_{k=1}^n 2^{-\nu/2} C_{s_k} + C_{s_0} \right]} \quad (11)$$

After applying the enhancement technique, an image with improved texture is obtained. [7].

## IV. EXPERIMENT

### A. Data preparation

The dataset comprises computed tomography (CT) scans presented as 12-bit images. Each pixel contains a value expressed in Hounsfield units (HU), as referenced in [13]. The Hounsfield unit (HU) is a linear transformation of the absorption coefficient ( $\mu$ ) of body tissue, as determined by means of tomographic measurement.

$$HU = 1000 \times \frac{\mu - \mu_B}{\mu_B - \mu_{\text{H}}}, \quad (12)$$

where  $\mu_B$  is represents the water absorption coefficient,  $\mu_{\text{H}}$  is air absorption coefficient.

When analyzing CT scans, a specific range of the image is chosen for examination. The range in question encompasses solely a specific portion of the Hounsfield scale, which is in alignment with the tissues of interest. The window's dimensions are defined by the centre position on the Hounsfield scale, which is referenced as WL (window level), and the width, which is referenced as WW (window width). To find the maximum and minimum values (referred to as  $I_{\min}$  and  $I_{\max}$ , respectively) within a given window, you can use the following formulas:

$$I_{\min} = WL - (WW \div 2) \quad (13)$$

$$I_{\max} = WL + (WW \div 2) \quad (14)$$

Any values below  $I_{\min}$  will appear black, in the resulting image, while any values above  $I_{\max}$  will appear white (Fig. 1).

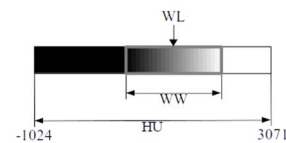


Fig. 1. Schematic of the Hounsfield scale window.

The image dataset was divided into discrete layers, to which a window (WL: 40, WW: 120) was applied for the



purpose of highlighting the brain tissue. The resulting images were saved in 8-bit Portable Network Graphics format. To guarantee the efficacy of the model development process, the dataset was divided into two distinct sets: a training set comprising 80% randomly selected images, and a test set comprising the remaining 20%.

The objective of this study is to develop a model that is capable of accurately segmenting intracranial hemorrhages, but it also extends beyond that. Accordingly, the objective was reduced to a single requirement: that the network should be capable of segmenting any type of haemorrhage present in the image. Furthermore, only images in which the haemorrhage was unquestionably present were considered, resulting in a total of 318 images in the dataset. It was also mentioned that masks based on fractional derivatives were studied in this work.

### B. Technologies used

The research was carried out utilizing the Python programming language. The image preprocessing was done using the OpenCV library of computer vision algorithms. To work with neural networks, the PyTorch framework [14] was used, which provides a convenient and flexible interface for parallel training of ANNs on GPUs using the CUDA technology. The MedicalTorch [15], a small PyTorch-based library containing popular ANN architectures and auxiliary functions for working with medical data, was used in this work too.

To train ANNs, powerful GPUs are required, which are accessible through many cloud services. For this research, the Google Colab cloud computing service was utilized. An Nvidia Tesla T4 with 16 GB of video memory was used at the time of the research.

### C. Using the PyTorch framework to train U-Net

An ICHDataset class was created, derived from PyTorch's standard Dataset class. Two objects of this class were made to represent the training and test subsets. The decision was taken to transfer complete images (512x512) to the network input, rather than dividing them into sections (160x160), as outlined in [3]. In order to reduce memory usage, the batch size was set to 1. Furthermore, a random affine transformation was incorporated into the training phase to enhance data diversity.

The neural network for segmentation utilized the standard U-Net architecture from the MedicalTorch library. Dropout and batch normalization parameters were configured at 0.4 and 0.1, respectively. The network weights were fine-tuned using the Adam optimizer with a learning rate of 0.001. The loss function used was the Dice coefficient with a negative sign, and its values ranged from 0 to -1. The value of the function was found to decrease during the training period.

The model was transferred to the GPU memory using PyTorch API [14]. The main training loop was implemented and repeated for 100 epochs. At each iteration, the training data batch was transferred to the GPU using the same method as for the model, and then passed to the network. The loss function value was calculated, followed by error backpropagation and network parameter optimization.

### D. Mask implementation based on Riesz fraction derivatives

The Python programming language was used to implement the algorithm described in [10]. The algorithm functions as a method that accepts an image, the Riesz derivative degree, and the window size as input. The resulting

output is an image with enhanced texture following the application of the FCD operator.

The function results in the creation of 8 W1 matrices with the coefficients calculated according to (8) using the NumPy module. The OpenCV library's convolution function is then utilized to apply each mask separately to the image in accordance with (9) and (10). The ultimate outcome is derived by aggregating the outcomes of applying individual masks according to (11).

The objective of this study is to examine the influence of the Riesz derivative-based operator on the quality of segmentation, a second version of the dataset was created. In this version, the operator was applied to all images with a mask size of 3x3 and a derivative order of  $v = 0.5$ .

## V. RESULTS AND DISCUSSIONS

### A. Analysis of segmentation solution results

During the training process, the neural network was trained to recognize and highlight hemorrhages in medical images. To evaluate its performance, 20% of images were set aside for testing and not used during training. Following each training session, the network was tested on a random image from this test set to see how well it was able to detect hemorrhages.

After the first epoch, the network was able to binarize the image, but it didn't highlight the areas that needed to be identified. It wasn't until the ninth epoch that the network began to successfully highlight hemorrhages, although they were still quite pronounced in the input image.

By the hundredth epoch, the network had significantly improved in detecting even subtle hemorrhages that were difficult to spot at first glance. Overall, the trained network was able to cope well with the assigned task of detecting hemorrhages in medical images.

Table III displays the test results of a network that underwent 100 epochs of training. In order to facilitate comparison, the table has been presented in the same format as Table I. The maximum and average values of both metrics for the network are higher than those reported in reference [3]. Moreover, the enhanced outcomes for image segmentation in cases of affected regions indicate the potential benefit of a combined model comprising a classification component and a segmentation component. Exploring such a model could be a promising area for future research on the topic presented in this paper.

TABLE III. ASSESSMENT OF SEGMENTATION ACCURACY

| Test dataset statistics | Dice coefficient | Jaccard coefficient |
|-------------------------|------------------|---------------------|
| Average                 | 0.452            | 0.353               |
| Minimum                 | 0.00             | 0.00                |
| Maximum                 | 0.876            | 0.780               |
| STD                     | 0.333            | 0.289               |

### B. Analysis of Image Quality Improvement through the Riesz Derivative

The objective of this study is to examine the applicability of the Riesz derivative-based operator to a specific dataset. Figure 2 illustrates the outcomes of utilising the FCD operator with varying window dimensions based on the Riesz derivative of order  $v = 0.5$  on an image extracted from the dataset. The smallest window size brings out more visible texture of the brain, while the contrast increases with larger

window sizes, but so does the noise. This aligns with the findings of the authors of the FCD operator which are shown in Table II. Therefore, it is advised that further research be conducted using smaller masks of 3x3 or 5x5.

Additionally, an investigation was conducted into the 7x7 FCD Operator based on the Riesz derivative of orders 0.1, 0.5 and 0.9. The result for the derivative of order 0.5 (Fig. 3) appears to be the most optimal visually, as enhancements in the texture are scarcely noticeable for order  $\nu = 0.1$ , while for  $\nu = 0.9$ , the presence of noise becomes increasingly discernible.

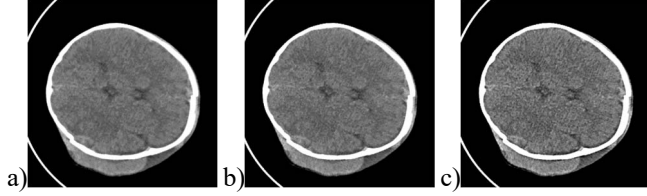


Fig. 2. The results of applying the FCD operator, which is based on the Riesz derivative of order  $\nu = 0.5$ , with different mask sizes are presented in the following figures. Figure (a) shows the original image, while figures (b) and (c) show the results of applying the operator to images with 3x3 and 7x7 masks, respectively.

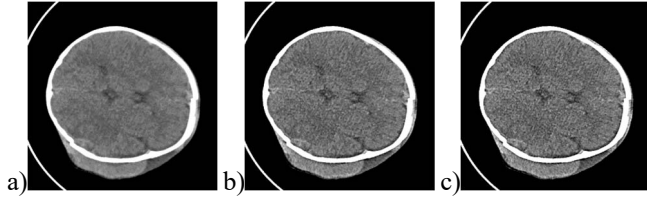


Fig. 3. The results demonstrate the implementation of the 7x7 FCD operator, based on the Riesz derivative of varying orders, on the input image (a), as well as on images with  $\nu = 0.5$  (b) and  $\nu = 0.9$  (c), respectively.

For an impartial assessment of the enhancement in image texture quality, a gray-level co-occurrence matrix (GLCM) approach, as suggested in [8], was utilized.

Given that the images under consideration may feature values ranging from 0 to 255, matrices sized 256x256 were generated. Additionally, as black pixel pairs ( $c_{0,0}$ ) in the dataset predominantly represent the background, and white pixel pairs ( $c_{255,255}$ ) correspond to bones, they were omitted from the texture analysis. This was achieved by assigning a value of 0 to the respective elements of the matrix.

Following the acquisition of four matrices, they were averaged to yield four characteristics: contrast (15), correlation (16), energy (17), and homogeneity (18). These calculations were conducted in accordance with the methodology outlined in reference [19].

$$Contrast = \sum_i \sum_j (i - j)^2 c_{i,j} \quad (15)$$

$$Correlation = \sum_i \sum_j \frac{(i - \mu_i)(j - \mu_j)c_{i,j}}{\sigma_i \sigma_j} \quad (16)$$

where  $\mu_i$ ,  $\mu_j$  represent the means and  $\sigma_i$ ,  $\sigma_j$  represent the standard deviations of the row and column sums of the GLCM matrix

$$Energy = \sum_i \sum_j c_{i,j}^2 \quad (17)$$

$$Homogeneity = \sum_i \sum_j \frac{c_{i,j}}{1 + |i - j|} \quad (18)$$

The following analysis was conducted on the original image and on images that were processed using the FCD operator with different parameters. In particular, the order  $\nu$  of the Riesz derivative was considered to have three values (0.9, 0.5, 0.1), and the mask size was considered to have three values (11x11, 7x7, 3x3). The average characteristics of all images in the dataset are shown in Fig. 4.

The analysis showed that increasing the order  $\nu$  of the Riesz derivative and window size leads to an increase in contrast and a decrease in correlation, energy, and homogeneity. Furthermore, it was observed that there is little effect of window size on correlation, contrast and homogeneity at lower values of  $\nu$

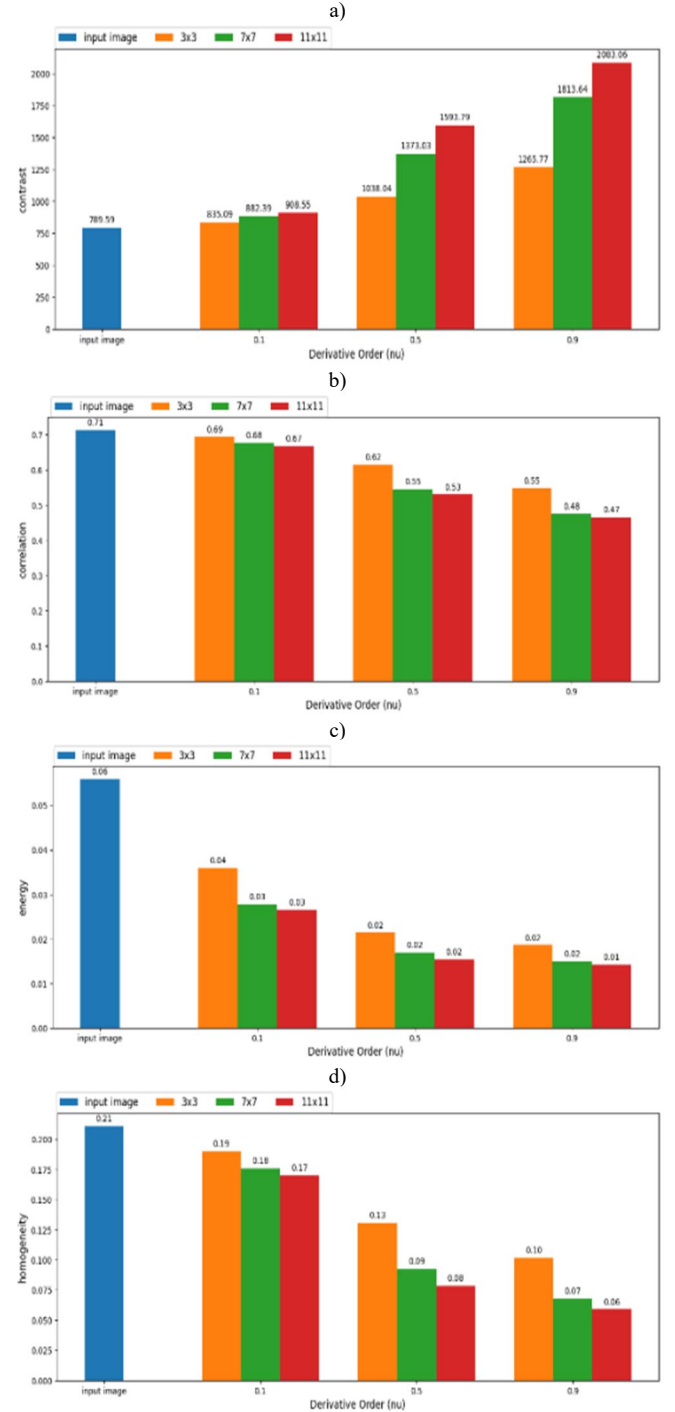


Fig. 4. Diagrams showing the main characteristics of GLCM for the studied data set for various parameters of the FCD operator: a) contrast; b) correlation; c) energy; d) homogeneity.

These results show a correlation with the findings from [8] for other fractional operators. An increase in contrast implies an increase in the difference between adjacent pixels, which can indicate an improved image clarity. A decrease in homogeneity indicates a decrease in the number of areas that have the same color tone. An increase in the complexity of the image texture is indicated by a decrease in correlation and energy. Energy values can be low because medical images tend not to have simple and predictable textures, which can also indicate increased noise. As a result, we decided to use the FCD operator, for the following experiment, with a mask size of  $3 \times 3$ , which is the smallest, and the order of the Riesz derivative  $\nu = 0.5$ .

### C. The Impact of the FCD Operator on Segmentation Quality

The objective was to ascertain whether utilising the FCD operator would prove beneficial in addressing the issue of intracranial haemorrhage segmentation, which had previously been identified as a challenge. In addition to assessing the enhancement in image texture, the entire training and test data sets were processed with the FCD operator using a  $3 \times 3$  window size based on the Riesz derivative of order  $\nu = 0.5$ . We then repeated the U-Net training and verification using the preprocessed data.

Table IV shows the results we obtained. We compared these results with those shown in Table III and concluded that using the FCD operator not only enhances the visual texture but also improves the quality of segmentation. The average value of the Dice coefficient for the test data set increased from 0.452 to 0.507 and the Jaccard coefficient increased from 0.353 to 0.398.

TABLE IV. EVALUATION OF THE ACCURACY OF THE SEGMENTATION OF THE DATA PROCESSED BY THE FCD OPERATOR

| Test dataset statistics | Dice coefficient | Jaccard coefficient |
|-------------------------|------------------|---------------------|
| Average                 | 0.507            | 0.398               |
| Minimum                 | 0.000            | 0.000               |
| Maximum                 | 0.921            | 0.854               |
| STD                     | 0.317            | 0.276               |

## VI. CONCLUSIONS

Techniques based on fractional order derivatives have exhibited promising results in enhancing medical image texture. However, the impact of such enhancements on applied problems in biomedical engineering has not been thoroughly studied. This paper investigates the influence of the Riesz fractional derivative based FCD operator on intracranial hemorrhage (ICH) segmentation.

Following a review of current developments in ICH segmentation, we selected an appropriate dataset with labeled hemorrhages. The U-Net network was trained on a portion of the dataset using a parallel algorithm based on the PyTorch framework, employing CUDA technology. Results were evaluated on another section of the dataset using Jaccard and Dice coefficients, achieving accuracies comparable to the baseline. Subsequently, the effect of the FCD operator on the selected data set was investigated, taking into account the order of the Riesz derivative ( $0 < \nu < 1$ ) and the window size. The texture quality of the processed image was quantitatively and visually assessed using GLCM. The optimal parameters were determined as a derivative of the Risa order of  $\nu = 0.5$  with a small window of size  $3 \times 3$  or  $5 \times 5$ . The dataset underwent preprocessing with the FCD operator of order 0.5 using a  $3 \times 3$

window, and the U-Net experiment was repeated. Results comparison revealed an improvement in accuracy by approximately 12%. While it is premature to discuss the automatic diagnosis of real patients, this study demonstrates the potential of this technology. With further improvements, a system based on artificial neural networks (ANN) could assist doctors in identifying problematic areas on a CT-scan. This work primarily focused on analyzing the impact of texture enhancement on ICH segmentation, thus investigating only the classic U-Net network. Considering more advanced ANN usage could be a potential direction for improvement in future studies.

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